

Strategic coexistence theory for evolutionary games, another case study

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So far, we have focused on simple deterministic 2×2 games. However, many ecological and evolutionary processes are intrinsically stochastic. Recently, Wang et al. (2025) showed that environmental fluctuations can promote coexistence in games that would otherwise lead to competitive exclusion in a deterministic environment. From the perspective of SCT, this suggests that stochasticity can stabilize competition by promoting the recovery of strategies when rare. Interestingly, Wang et al. (2025) also showed that stochasticity can generate bistable coexistence and multiple internal equilibrium. Thus, although stochasticity can stabilize competition near the boundaries, it can also generate destabilizing frequency dependence in the interior. This immediately raises a question of stochasticity can generate sufficiently strong positive frequency dependence to destabilize competition, thus pushing the system out of the coexistence regime. Here, we use SCT to separate these effects and clarify whether stochasticity stabilizes or destabilizes strategic competition.

Following Wang et al. (2025), we consider a 2×2 game in the presence of environmental noise. The stochastic payoff matrix is given by

$$\begin{pmatrix} R & S \\ T & P \end{pmatrix} + \frac{\zeta}{\sqrt{s}} \begin{pmatrix} \sigma_R & \sigma_S \\ \sigma_T & \sigma_P \end{pmatrix}, \quad (1)$$

where the first matrix represents deterministic payoffs, ζ represents a noise term with zero mean and unit variance, and the second matrix determines the intensity of environmental stochasticity. Assuming weak selection $s \ll 1$, the evolution of two competing strategies in a large population can be described by the following equation:

$$\frac{dx}{dt} = sx(1-x) \left(\pi_C - \pi_D + \left(\frac{1}{2} - x \right) (\sigma_C - \sigma_D)^2 \right), \quad (2)$$

where $\pi_C = Rx + S(1-x)$ and $\pi_D = Tx + P(1-x)$ and $\sigma_C = \sigma_R x + \sigma_S(1-x)$ and $\sigma_D = \sigma_T x + \sigma_P(1-x)$. For this system, the invasion growth rates of each strategy when rare can be written as:

$$r_C = S - P + \frac{1}{2}(\sigma_S - \sigma_P)^2, \quad (3)$$

$$r_D = T - R + \frac{1}{2}(\sigma_T - \sigma_R)^2. \quad (4)$$

32 This shows that environmental stochasticity always increases the invasion growth
 33 rates, meaning that it is intrinsically stabilizing.

34 Then, an extended SCT decomposition shows that niche overlap and fitness dif-
 35 ference can be written as:

$$\rho = \exp \left(\frac{R + P - S - T}{2} - \frac{(\sigma_R - \sigma_T)^2 + (\sigma_S - \sigma_P)^2}{4} \right) \quad (5)$$

$$\frac{\kappa_D}{\kappa_C} = \exp \left(\frac{T + P - R - S}{2} + \frac{(\sigma_R - \sigma_T)^2 - (\sigma_S - \sigma_P)^2}{4} \right) \quad (6)$$

36 Writing $\hat{\rho}$ and $\hat{\kappa}_D/\hat{\kappa}_C$ to denote niche overlap and fitness difference in the absence of
 37 stochasticity and taking logs of both sides for convenience, we have:

$$\Delta \log \rho = \log \rho - \log \hat{\rho} = - \frac{(\sigma_R - \sigma_T)^2 + (\sigma_S - \sigma_P)^2}{4} \quad (7)$$

$$\Delta \log \frac{\kappa_D}{\kappa_C} = \log \frac{\kappa_D}{\kappa_C} - \log \frac{\hat{\kappa}_D}{\hat{\kappa}_C} = \frac{(\sigma_R - \sigma_T)^2 - (\sigma_S - \sigma_P)^2}{4}. \quad (8)$$

38 Therefore, stochasticity always reduces niche overlap and therefore increases niche
 39 difference. In contrast, stochasticity can either amplify or reduce fitness differences
 40 depending on which strategy's invasion rate is more strongly affected by environ-
 41 mental fluctuations.

42 Notably, changes in fitness difference driven by environmental stochasticity will
 43 be always bounded by those in niche overlap:

$$\Delta \log \rho \leq \Delta \log \frac{\kappa_D}{\kappa_C} \leq -\Delta \log \rho \quad (9)$$

44 This result implies that environmental stochasticity cannot generate sufficient fitness
 45 difference to push the system out of the coexistence regime.

46 This example also illustrates an important limitation of SCT, as well as the
 47 original MCT. Since SCT and MCT solely depend on boundary conditions, it cannot
 48 predict the dynamics within interior, especially the presence of multiple internal
 49 equilibrium. For example,

50 Supp

51 Consider a competition of two strategies under stochastic payoffs:

$$\begin{pmatrix} R & S \\ T & P \end{pmatrix} + \frac{\zeta}{\sqrt{s}} \begin{pmatrix} \sigma_R & \sigma_S \\ \sigma_T & \sigma_P \end{pmatrix}, \quad (10)$$

52 Assuming weak selection $s \ll 1$, the evolution of two competing strategies in a large
 53 population can be described by the following equation:

$$\frac{dx}{dt} = sx(1-x) \left(\pi_C - \pi_D + \left(\frac{1}{2} - x \right) (\sigma_C - \sigma_D)^2 \right), \quad (11)$$

54 To simplify, we write

$$\frac{dx}{dt} = sx(1-x)G(x), \quad (12)$$

55 where

$$G(x) = \left(\pi_C - \pi_D + \left(\frac{1}{2} - x \right) (\sigma_C - \sigma_D)^2 \right). \quad (13)$$

56 Thus, the sign of $G(x)$ at boundary conditions determines whether a strategy can
57 invade when the competing strategy is at equilibrium.

58 Note that we cannot immediately apply the original SCT presented in the main
59 text here because $G(x)$ does not uniquely separate into two payoff measures. In-
60 stead, we can still utilize the invasion growth rate when rare and follow the same
61 procedure. In particular, the mutual invasion condition at monomorphic boundary
62 can be written as:

$$0 < G(0) \quad (14)$$

$$0 < -G(1) \quad (15)$$

63 Exponentiating both, we can combine them into a single inequality:

$$\exp(G(1)) < 1 < \exp(G(0)) \quad (16)$$

64 Dividing both sides by $\exp((G(1) + G(0))/2)$, we get:

$$\exp\left(\frac{G(1) - G(0)}{2}\right) < \exp\left(-\frac{G(1) + G(0)}{2}\right) < \exp\left(\frac{G(0) - G(1)}{2}\right), \quad (17)$$

65 which gives us expressions for niche overlap and fitness difference:

$$\rho = \exp\left(\frac{G(1) - G(0)}{2}\right) \quad (18)$$

$$\frac{\kappa_D}{\kappa_C} = \exp\left(-\frac{G(1) + G(0)}{2}\right) \quad (19)$$

66 Then, by substituting expressions for $G(0)$ and $G(1)$, we obtain following expressions
67 for niche and fitness difference:

$$1 - \rho = 1 - \exp\left(\frac{R + P - S - T}{2} - \frac{(\sigma_R - \sigma_T)^2 + (\sigma_S - \sigma_P)^2}{4}\right) \quad (20)$$

$$\frac{\kappa_D}{\kappa_C} = \exp\left(\frac{T + P - R - S}{2} + \frac{(\sigma_R - \sigma_T)^2 - (\sigma_S - \sigma_P)^2}{4}\right) \quad (21)$$

68 Now, to study the interior behavior of the system, we assume that the amount
69 of stochasticity is proportional to deterministic payoffs (Wang et al., 2025):

$$\begin{pmatrix} \sigma_R & \sigma_S \\ \sigma_T & \sigma_P \end{pmatrix} = k \begin{pmatrix} R & S \\ T & P \end{pmatrix}. \quad (22)$$

70 Then, the expression for $G(x)$ can be written as:

$$G(x) = ((R - T)x + (S - P)(1 - x)) \times \quad (23)$$

$$\left(1 + k^2 \left(\frac{1}{2} - x\right) ((R - T)x + (S - P)(1 - x))\right) \quad (24)$$

$$= ((R + P - T - S)x + (S - P)) \times \quad (25)$$

$$\left(1 + k^2 \left(\frac{1}{2} - x\right) ((R + P - T - S)x + (S - P))\right) \quad (26)$$

$$= k^2(R + P - T - S)^2(x - x^*) \left(K + \left(\frac{1}{2} - x\right)(x - x^*)\right), \quad (27)$$

71 where $x^* = (P - S)/(R + P - T - S)$ and $K = 1/(k^2(R + P - T - S))$. Therefore,
72 this system has interal equilibrium $x = x^*$ when $P - S > 0$ and $R - T > 0$. It can
73 also have two additional equilibrium when $K + \left(\frac{1}{2} - x\right)(x - x^*) = 0$ has positive,
74 real roots. In particular, the latter equation can be written as:

$$x^2 - \left(\frac{1}{2} + x^*\right)x + \frac{x^*}{1} - K = 0. \quad (28)$$

75 Thus, two additional equilibrium correspond to:

$$x = \frac{x^* + \frac{1}{2} \pm \sqrt{\left(\frac{1}{2} + x^*\right)^2 + 4K}}{2}. \quad (29)$$

76 This shows that two internal equilibrium exists when

$$\left(\frac{1}{2} + x^*\right)^2 + 4K > 0. \quad (30)$$

77 References

78 Wang, G., Q. Su, L. Wang, and J. B. Plotkin (2025). Evolution of social behaviors
79 in noisy environments. *arXiv*.